Algorithms and Data Structures 2.

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Breadth First Search – BFS

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Practice 3

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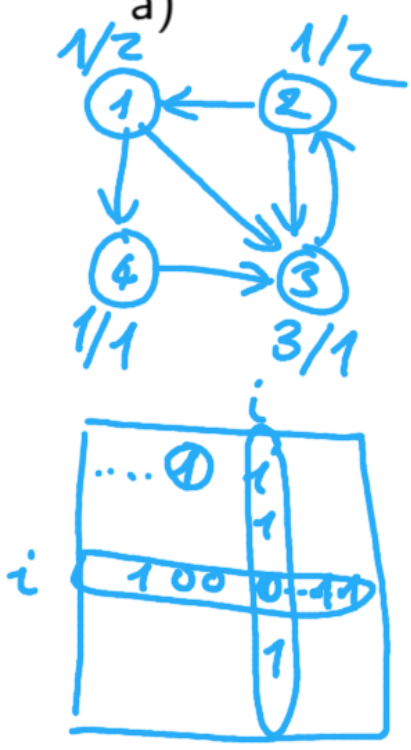
Exercise

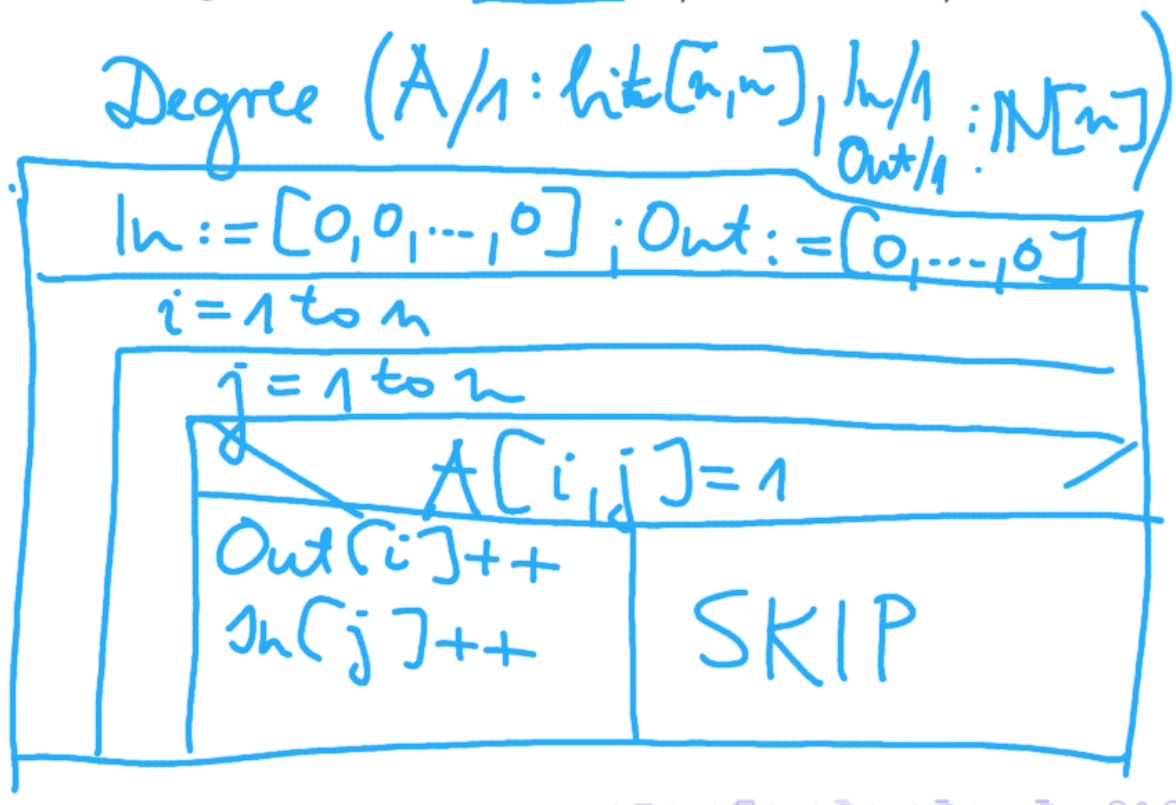
Design an algorithm which calculates the in-degrees and out-degrees of all the vertices of a graph given in (a) adjacency matrix representation or (b) adjacency list representation. These values should be given in two arrays $In/1$ and $Out/1$.
a)

Exercise

Design an algorithm which calculates the in-degrees and out-degrees of all the vertices of a graph given in (a) adjacency matrix representation or (b) adjacency list representation. These values should be given in two arrays $In/1$ and $Out/1$.

a)





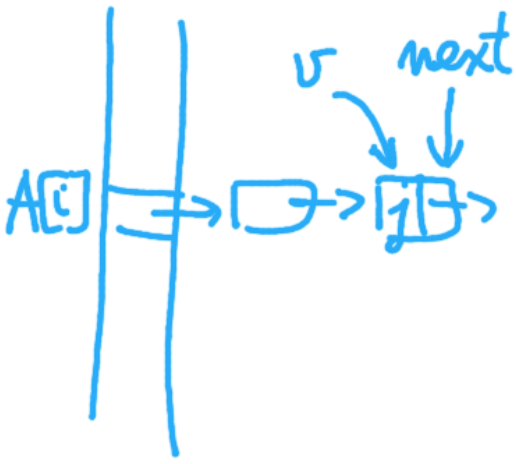
Exercise

Design an algorithm which calculates the in-degrees and out-degrees of all the vertices of a graph given in (a) adjacency matrix representation or (b) adjacency list representation. These values should be given in two arrays $In/1$ and $Out/1$.
b)

Exercise

Design an algorithm which calculates the in-degrees and out-degrees of all the vertices of a graph given in (a) adjacency matrix representation or (b) adjacency list representation. These values should be given in two arrays $In/1$ and $Out/1$.

b)



Degree ($A/_1$: Edge*[n], $In/_1$, $Out/_1$: IN[n])

In := [0, ..., 0], Out := [0, ..., 0]

i = 1 to n

p := A[i]

p ≠ Ø

Out[i]++, In[p→v]++

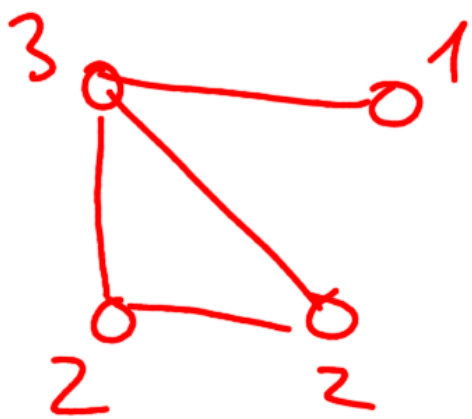
p := p→next

Exercise

directed

Design an algorithm which calculates the in-degrees and out-degrees of all the vertices of a* graph given in (a) adjacency matrix representation or (b) adjacency list representation. These values should be given in two arrays $In/1$ and $Out/1$.
b)

Undirected?



degree

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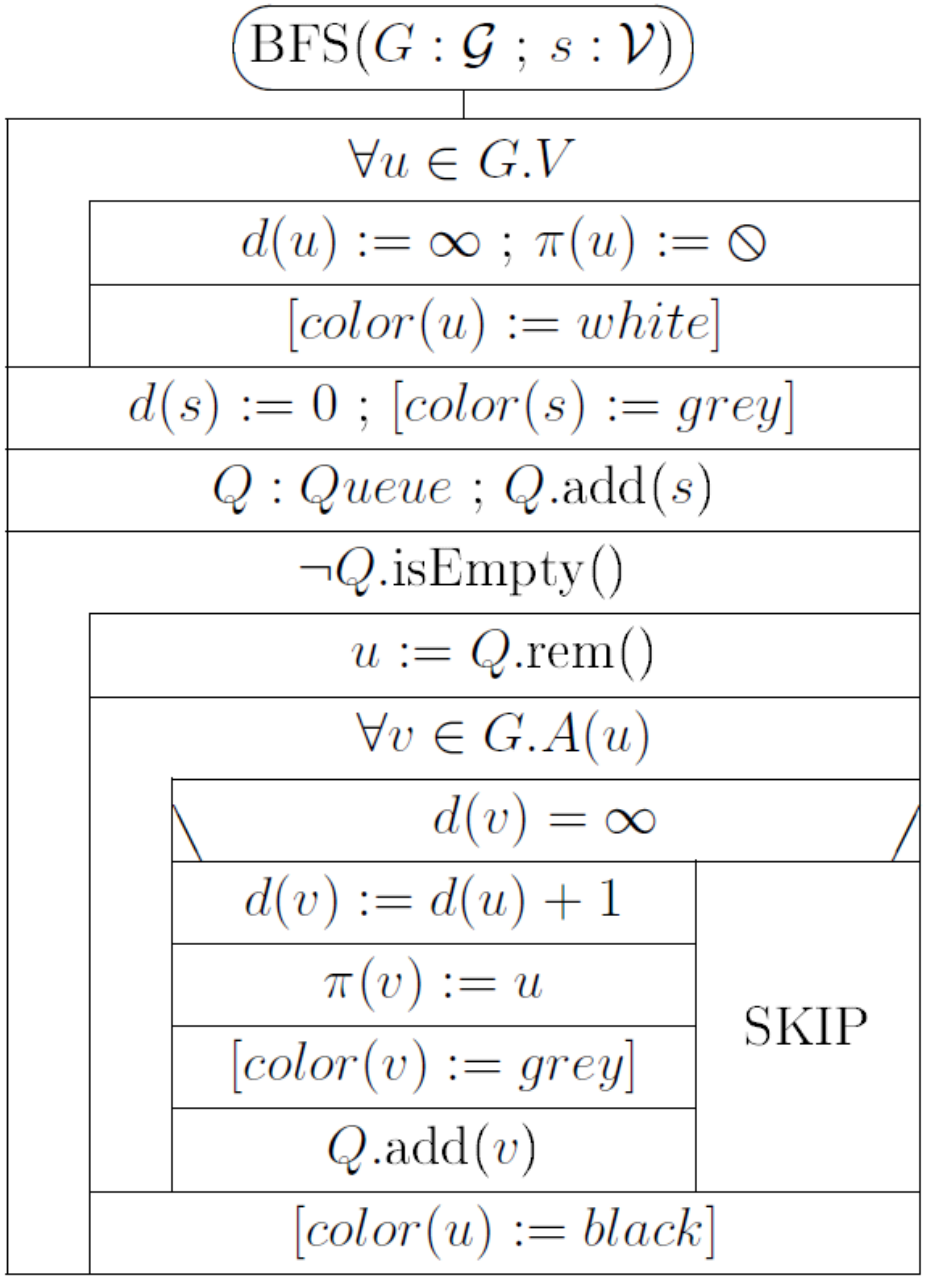
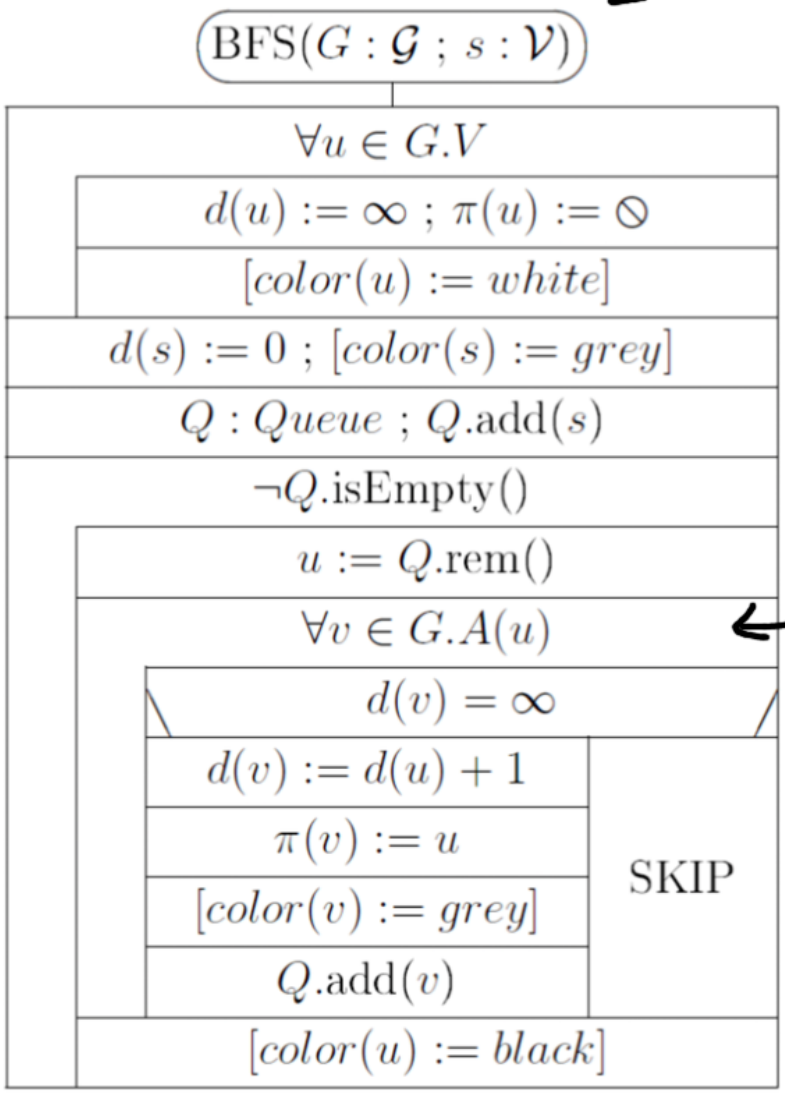
Breadth-first search

- Breadth-first search (or BFS) will be considered on both undirected or directed graphs.
- We will fix a source vertex $s \in G.V$.
- We will find the shortest path to each other vertex available from s.
- If there are more than one shortest paths, then we only compute one of them.

Breadth-first search



Breadth-first search



abstract graph repr.

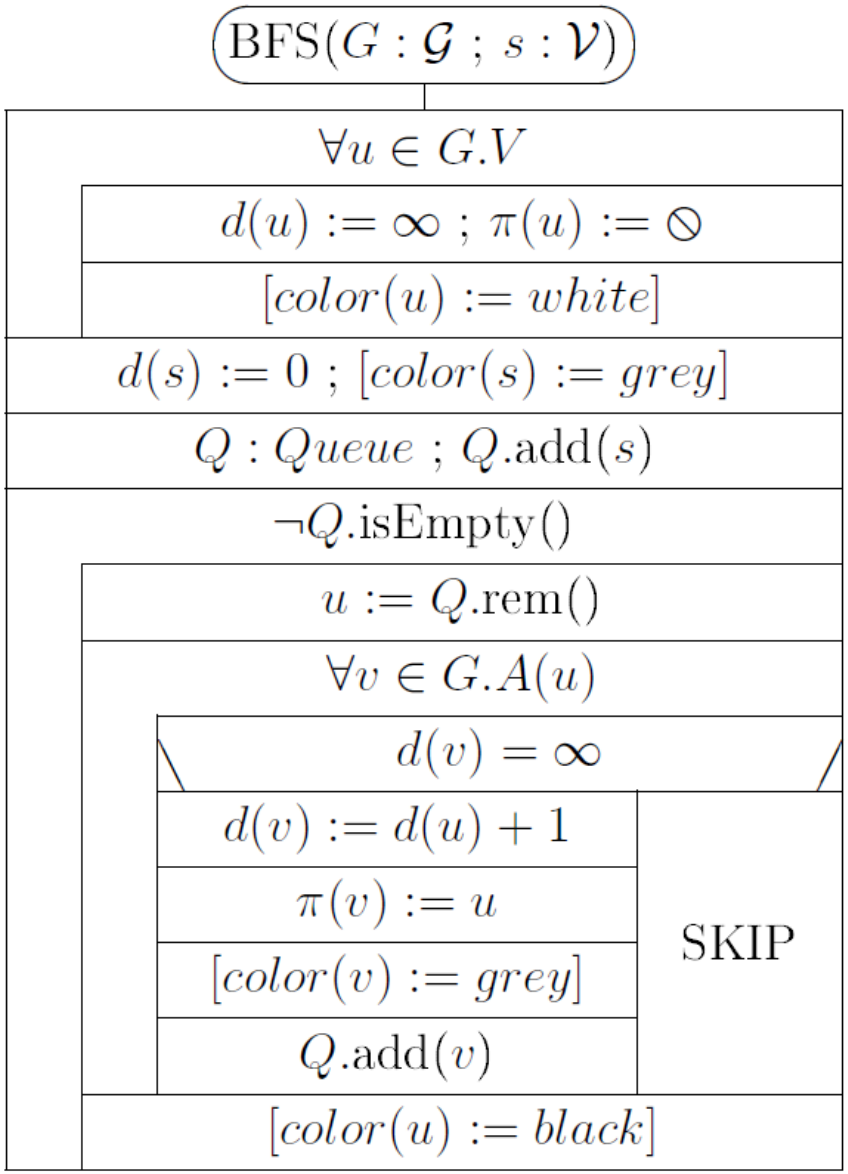
initialization

for all children/neighbors of u

Breadth-first search

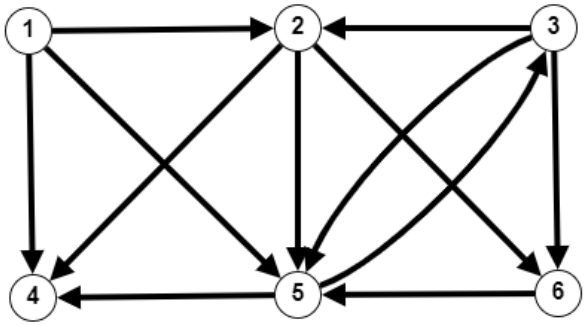
- For every vertex u we will have three labels:
  - $d(u)$ the length of the shortest $s \rightsquigarrow u$ path found
    $d(u) = \infty$ means no such path has been found so far
    $d(u) = 0$ is only true for $u = s$
  - $\pi(u)$ the parent vertex of u on the $s \rightsquigarrow u$ path found.
    $\pi(u) = \oslash$ means that we have not found such path or $u = s$
  - $color(u)$ can be omitted from the program (useful for illustration)
    - *white* if it has not been found yet
    - *grey* if it has been found but not yet processed
    - *black* if it has already been processed
- In the first loop of the algorithm these labels are set:
  $\forall u \in G.V$: $d(u) = \infty$, $\pi(u) = \oslash$ and $color(u) = white$
  for s source: $d(s) = 0$, $\pi(s) = \oslash$, $color(s) = grey$

Breadth-first search



- s is added to queue Q
- Main loop: in every iteration a vertex u is removed from Q
- and "expanded", i.e. all its neighbours are considered, and if they have not been found before ($d(v) = \infty$) then their d, π and *color* labels are set, and they are added to Q
- Expanded vertices become black.

Example for BFS



$\neg Q.\text{isEmpty}()$

$u := Q.\text{rem}()$

$\forall v \in G.A(u)$

$d(v) = \infty$

$d(v) := d(u) + 1$

$\pi(v) := u$

$[color(v) := grey]$

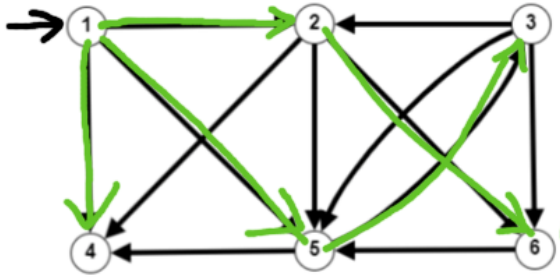
$Q.\text{add}(v)$

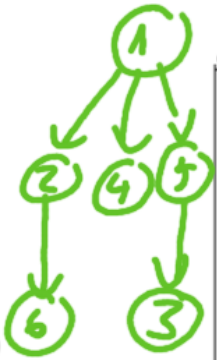
SKIP

$[color(u) := black]$

changes of d						ex- panded vertex : d	Q : Queue	changes of π					
1	2	3	4	5	6			1	2	3	4	5	6
							$\langle \quad \rangle$						
						final d and π values							

Example for BFS





BFS-tree

initialize

$\neg Q.\text{isEmpty}()$

$u := Q.\text{rem}()$

$\forall v \in G.A(u)$

$d(v) = \infty$

$d(v) := d(u) + 1$

$\pi(v) := u$

$[color(v) := grey]$

$Q.\text{add}(v)$

SKIP

$[color(u) := black]$

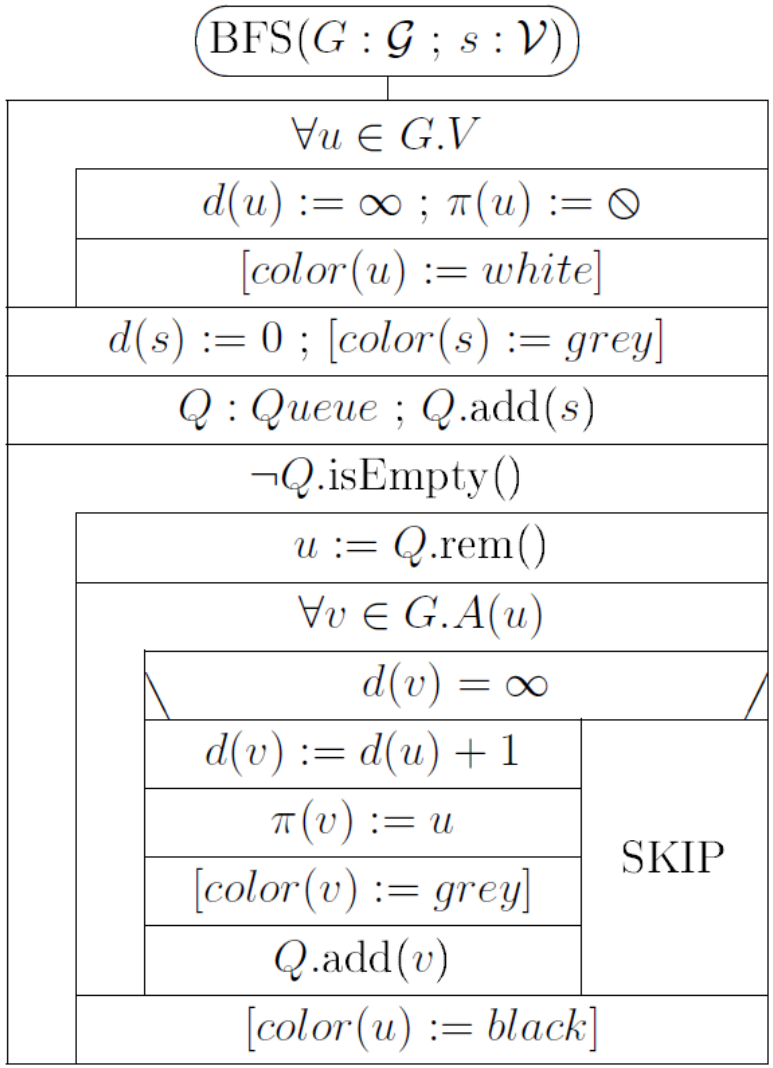
$(\pi(u), u)$

changes of d						ex-	Q :	changes of π					
1	2	3	4	5	6	panded	Queue	1	2	3	4	5	6
0	∞	∞	∞	∞	∞	vertex : d	⟨1⟩	0	0	0	0	0	0
0	1	∞	1	1	∞	1 : 0	⟨2,4,5⟩	0	1	0	1	1	0
0	1	∞	1	1	2	2 : 1	⟨4,5,6⟩	0	1	0	1	1	2
						4 : 1	⟨5,6⟩						
		2				5 : 1	⟨6,3⟩			5			
						6 : 2	⟨3⟩						
						3 : 2	⟨⟩						
0	1	2	1	1	2	final d and π values		0	1	5	1	1	2

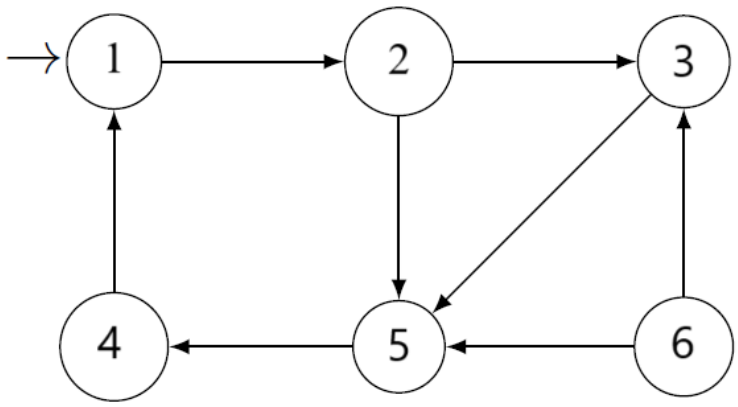
Properties

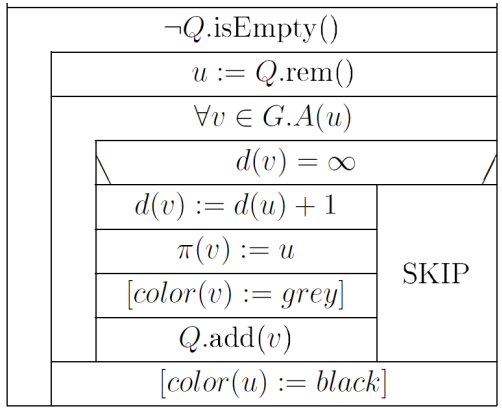
- Edges $(\pi(u), u)$ form the **breadth-first tree** or **shortest paths tree** with source vertex s of G.
- The set of vertices with d value k are the vertices on level k of the breadth-first tree. These are the vertices of G at distance k from source vertex s.
- During the algorithm when a vertex v with value $d(v) = k$ is being expanded (or processed), then all the vertices in the queue Q have d value k or $k + 1$, and the ones with d value k come first.
- When the last vertex with d value k has been expanded, then the set of vertices in Q are exactly the set of vertices with d value $k + 1$, i.e. the vertices at distance $k + 1$ from s.

Properties of Breadth-first search



Example for BFS





changes of d						ex- panded vertex : d	Q : Queue	changes of π					
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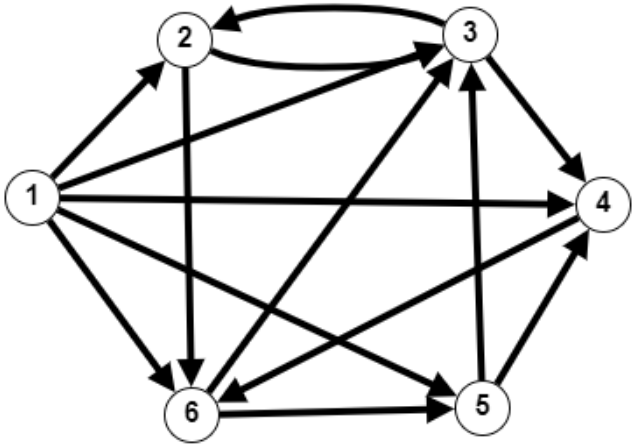
							$\langle \quad \rangle$						
						final d and π values							

Example for BFS



changes of d						ex- panded vertex : d	Q : Queue	changes of π					
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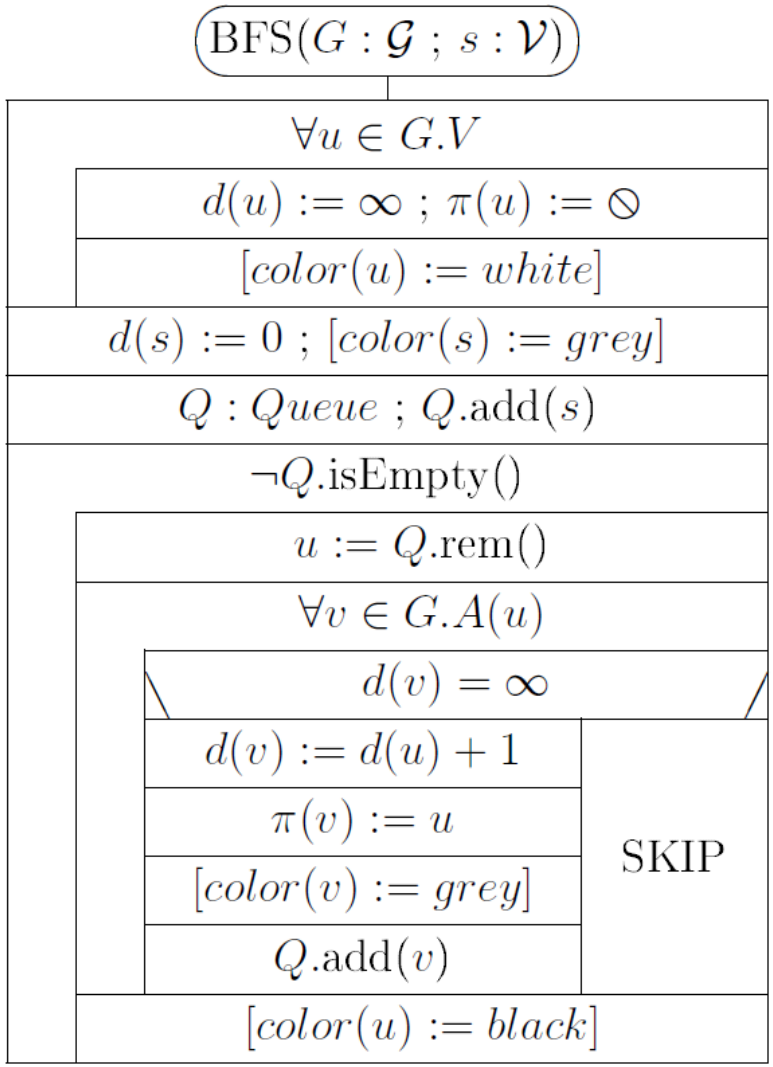
						final d and π values							

Optional assignment

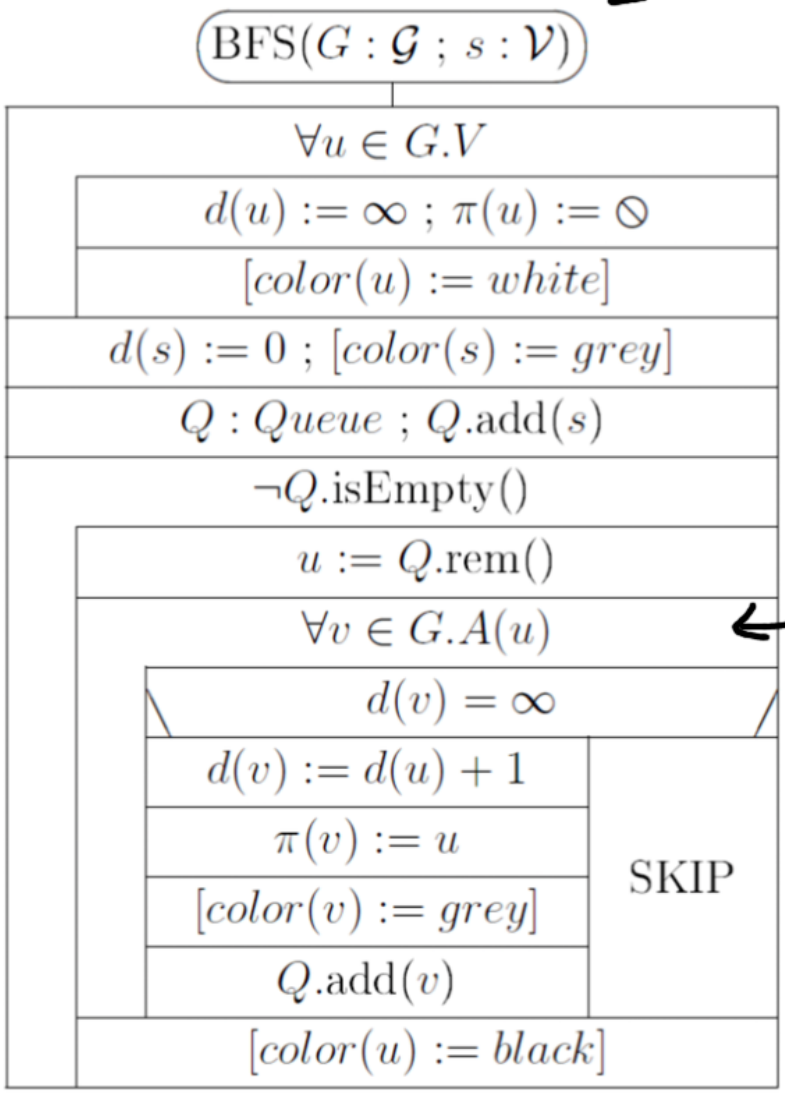
Let us suppose that we have run precedure $BFS(G, s)$. Write an algorithm $printShortestPathTo(v)$ which prints the shortest path $s \rightsquigarrow v$ that has been calculated by BFS if v is available from vertex s, and prints *"v is not available from s"* otherwise. Both recursive and non recursive algorithms can be given to solve this problem.

Efficiency of BFS



- $n = |G.V|$ and $m = |G.E|$
- Initializing loop iterates n times.
- The main loop iterates as many times as the number of vertices available from s (which is ≥ 1 and $\leq n$).
- For the inner loop the max number of iterations is m or $2m$ (directed, undirected), while the min is 0.
- Time complexity: $MT(n, m) \in \Theta(n + m)$, $mT(n, m) \in \Theta(n)$

Breadth-first search



abstract graph repr.

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Transitive closure

For each pair of vertices (u, v) of a network/graph, we want to determine whether there is any path form u to v or not.

We are interested neither in the path nor in its length. For this purpose we introduce the following notion.

Definition: Given graph $G = (V, E)$ its **transitive closure** is relation $T \subseteq V \times V$ where:

$(u, v) \in T \iff \exists$ path from vertex u to vertex v in graph G.

Transitive closure

$(u, v) \in T \iff \exists$ path from vertex u to vertex v in graph G.

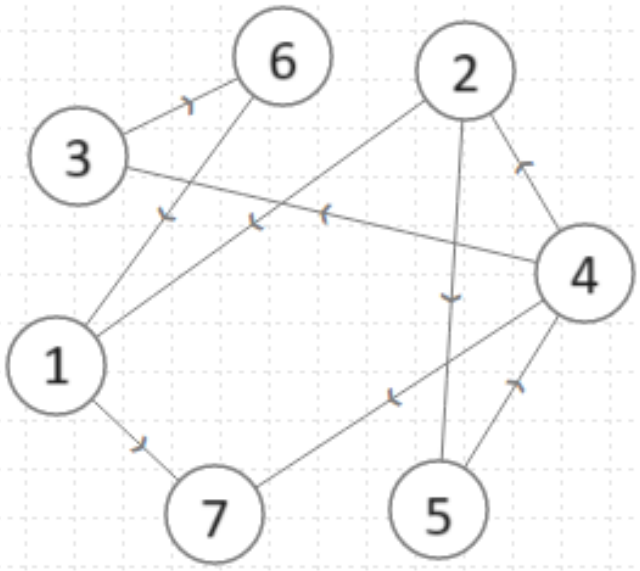
If the vertices of the graph are represented by indeces $1, 2, \ldots, n$, then we can represent the transitive closure of G with matrix

$$T/1 : \mathbb{B}[n, n]$$

where

$$T[i, j] = 1 \iff \exists \text{ path } i \rightsquigarrow j \text{ in the graph.}$$

Example



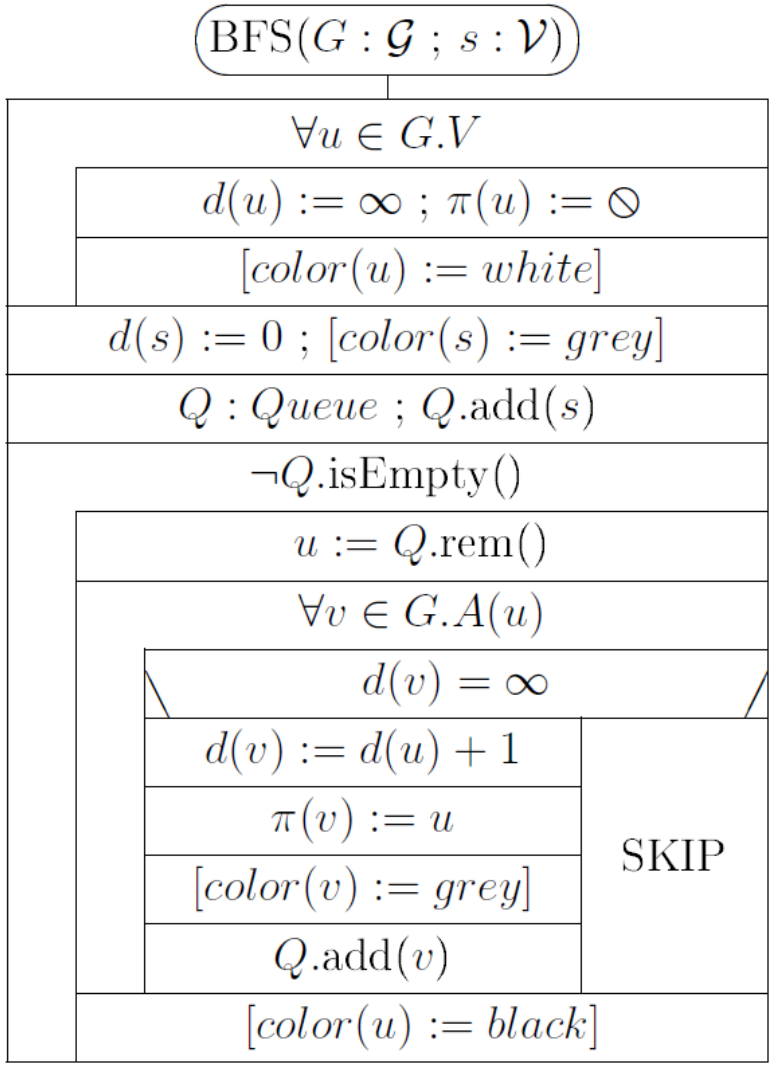
Directed Graph

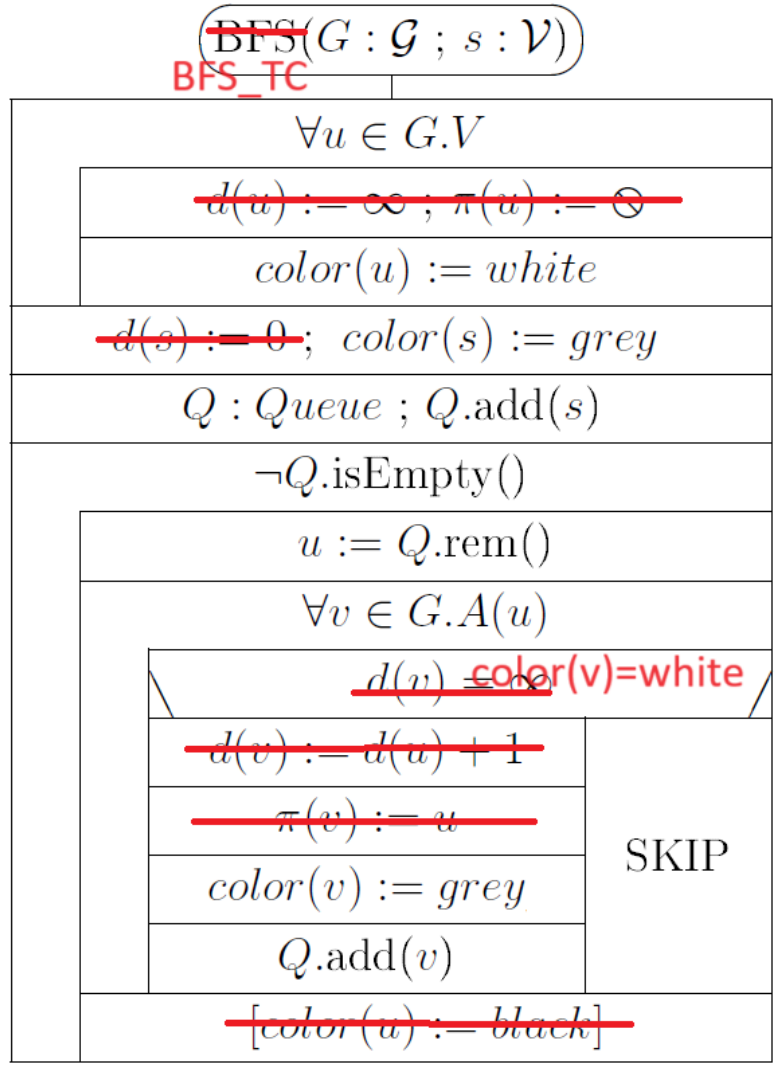
	1	2	3	4	5	6	7
1	1	0	0	0	0	0	1
2	1	1	1	1	1	1	1
3	1	0	1	0	0	1	1
4	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1
6	1	0	0	0	0	1	1
7	0	0	0	0	0	0	1

Transitive Closure

How can we calculate this matrix T?

First solution: perform a BFS from each vertex as source. We do not need to compute distances, nor parent vertices, but only reachability. In BFS a vertex is reached if it is not white.

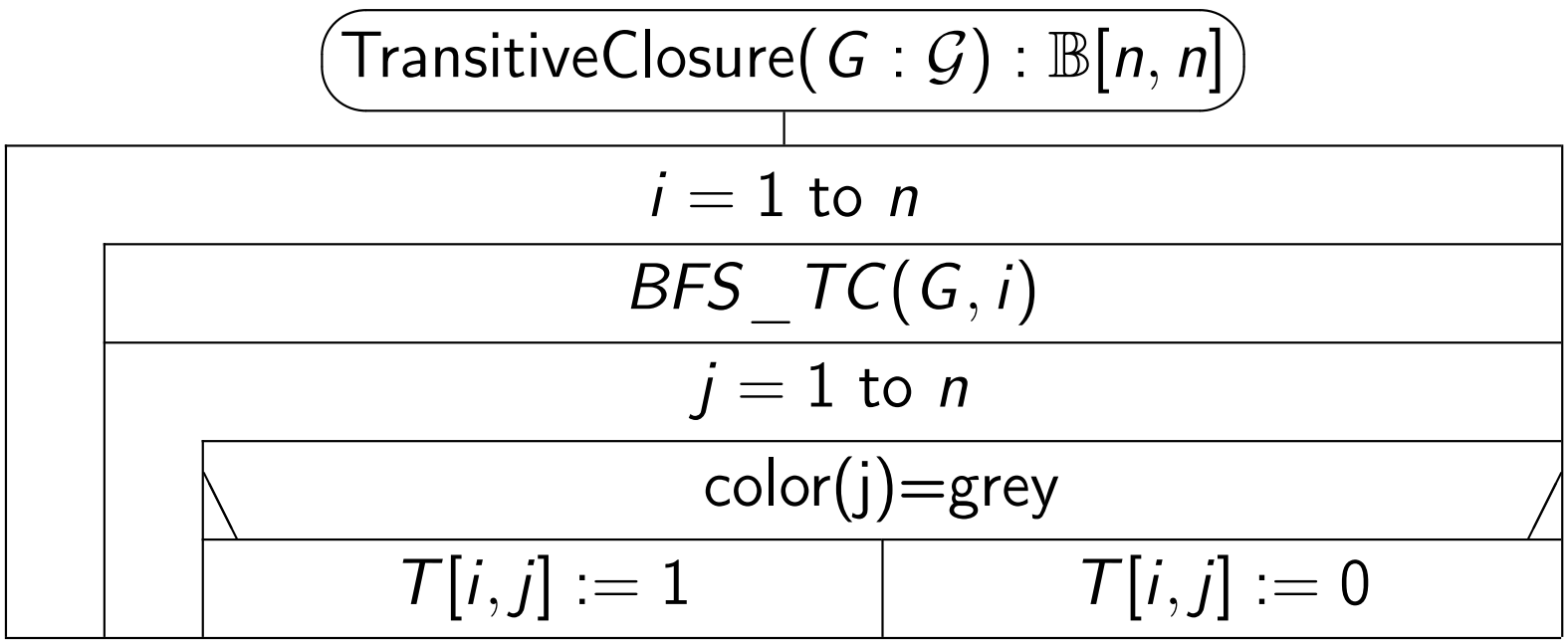




After running algorithm BFS_TC with source vertex i we have

$$T[i,j] = 1 \iff \text{color(j)=grey}$$

We obtain row i of matrix T.
If we run this algorithm for all vertices i, then we obtain T.



The time complexity will then be n times $\Theta(m+n)$, which is

$$\Theta(n(n+m))$$

For sparse graphs this is $\Theta(n^2)$, but for dense graphs it is $\Theta(n^3)$.

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$$\Theta(n(n+m))$$

For sparse graphs this is $\Theta(n^2)$, but for dense graphs it is $\Theta(n^3)$.

$m \in O(n)$ $\quad$ $m \in \Omega(n^2)$

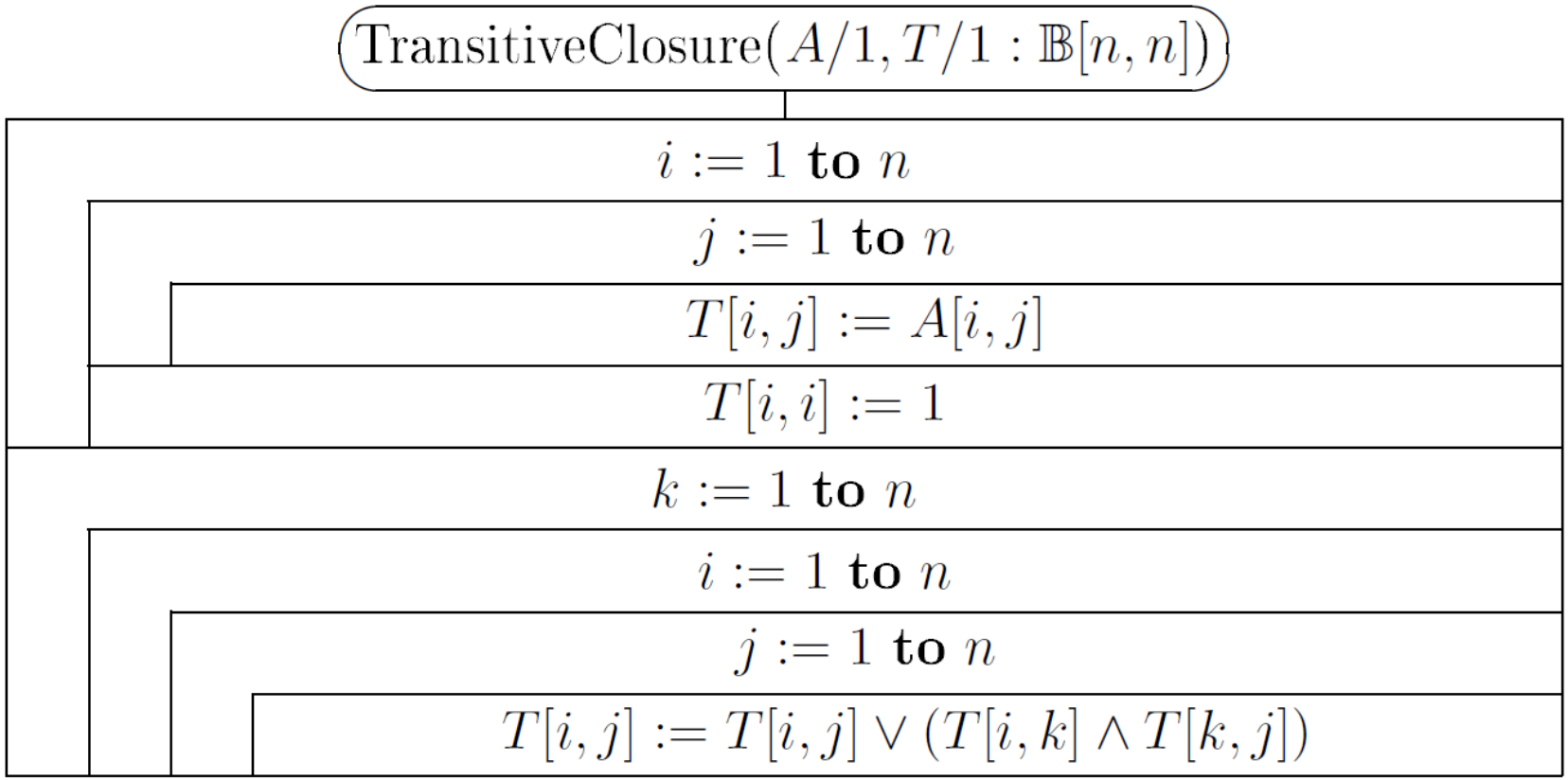
Another way of calculating transitive closure:

Suppose the graph is represented with adjacency matrix

$$A/1 : \mathbb{B}[n, n].$$

Now $\mathbb{B} = \{0, 1\}$ and we will use Boolean operators "AND" and "OR" (as $\wedge$ and $\vee$) on this set.

Another way of calculating transitive closure:

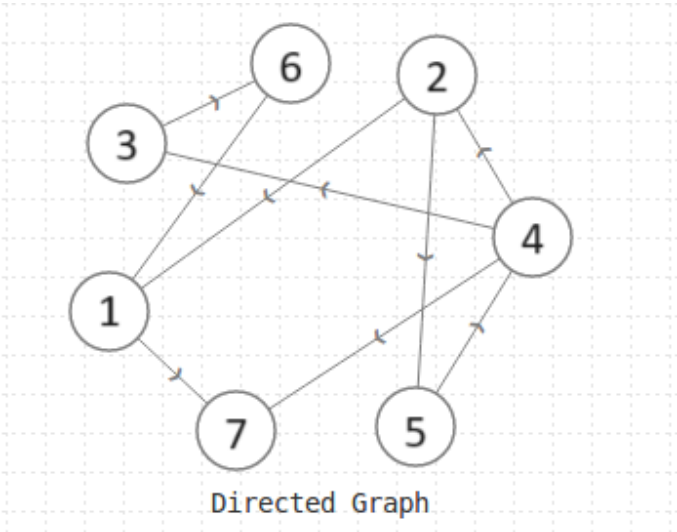


Clearly the time complexity for this algorithm is $\in \Theta(n^3)$.

Why is this algorithm good for computing Transitive Closure?

Notation. Let $i \overset{k}{\rightsquigarrow} j$ denote a path from vertex i to vertex j which uses only vertices $\leq k$ as inner vertices.
($i > k$ and/or $j > k$ is possible.)

So the path can be $< i, u_1, u_2 \ldots, u_n, j >$ where $u_1, u_2, \ldots, u_n \leq k$.



In this graph there is no $2 \overset{1}{\rightsquigarrow} 3$ path, nor $2 \overset{4}{\rightsquigarrow} 3$ or $2 \overset{0}{\rightsquigarrow} 3$ paths, but there is $2 \overset{5}{\rightsquigarrow} 3$ path. ($i \overset{0}{\rightsquigarrow} j$ is a path that has no inner vertices, i.e. it is an edge.) There is no $3 \overset{5}{\rightsquigarrow} 1$ path, but there is $3 \overset{6}{\rightsquigarrow} 1$ path.

We will build a series of matrices: $T^{(0)}$, $T^{(1)}$, $T^{(2)}$...
$T^{(n)} \in \mathbb{B}[n, n]$ with the following definition:

Definition.

$$T^{(k)}[i, j] = 1 \iff \exists\ i \overset{k}{\rightsquigarrow} j$$

where $k \in 0...n$ and $i, j \in 1..n$

For every $k = 0, 1, ..., n$ we will compute matrix $T^{(k)}$, and clearly

$$T^{(n)} = T.$$

Recursive relation among the T matrices:

- $T^{(0)} = A + I$
  - So $T^{(0)}$ is almost the same as the adj. matrix, but the main diagonal is changed to all 1's.

- $T^{(k)}[i,j] = T^{(k-1)}[i,j] \vee (T^{(k-1)}[i,k] \wedge T^{(k-1)}[k,j])$
  - There is a $i \overset{k}{\rightsquigarrow} j$ path if and only if
    - either there is an $i \overset{k-1}{\rightsquigarrow} j$ path
    - or there are paths $i \overset{k-1}{\rightsquigarrow} k$ and $k \overset{k-1}{\rightsquigarrow} j$

Recursive relation among the T matrices:

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    - or there are paths $i \overset{k-1}{\rightsquigarrow} k$ and $k \overset{k-1}{\rightsquigarrow} j$

$\{1, 2, \ldots, k\}$

$\{1, 2, \ldots, k-1\}$

Use these properties to prove that the following algorithm produces the matrix $T = T^{(n)}$.

```
i := 1 to n
    j := 1 to n
        T^(0)[i, j] := A[i, j]
    T^(0)[i, i] := 1
k := 1 to n
    i := 1 to n
        j := 1 to n
            T^(k)[i, j] := T^(k-1)[i, j] ∨ (T^(k-1)[i, k] ∧ T^(k-1)[k, j])
```

$T^{(0)}[i,j]: \begin{cases} 1 & \exists\, i \to j \text{ no other vertices} \\ 0 & \nexists \text{ --''--} \end{cases}$

Use these properties to prove that the following algorithm produces the matrix $T = T^{(n)}$.

$i := 1$ **to** n

  $j := 1$ **to** n

    $T^{(0)}[i,j] := A[i,j]$

  $T^{(0)}[i,i] := 1$

$k := 1$ **to** n

  $i := 1$ **to** n

    $j := 1$ **to** n

      $T^{(k)}[i,j] := T^{(k-1)}[i,j] \vee (T^{(k-1)}[i,k] \wedge T^{(k-1)}[k,j])$

$T^{(0)} \to T^{(1)} \to T^{(2)} \to T^{(3)} \to \ldots$

Do we really need all the $T^{(k)}$ matrices? Space complexity is $\in \Theta(n^3)$.

No, we don't.

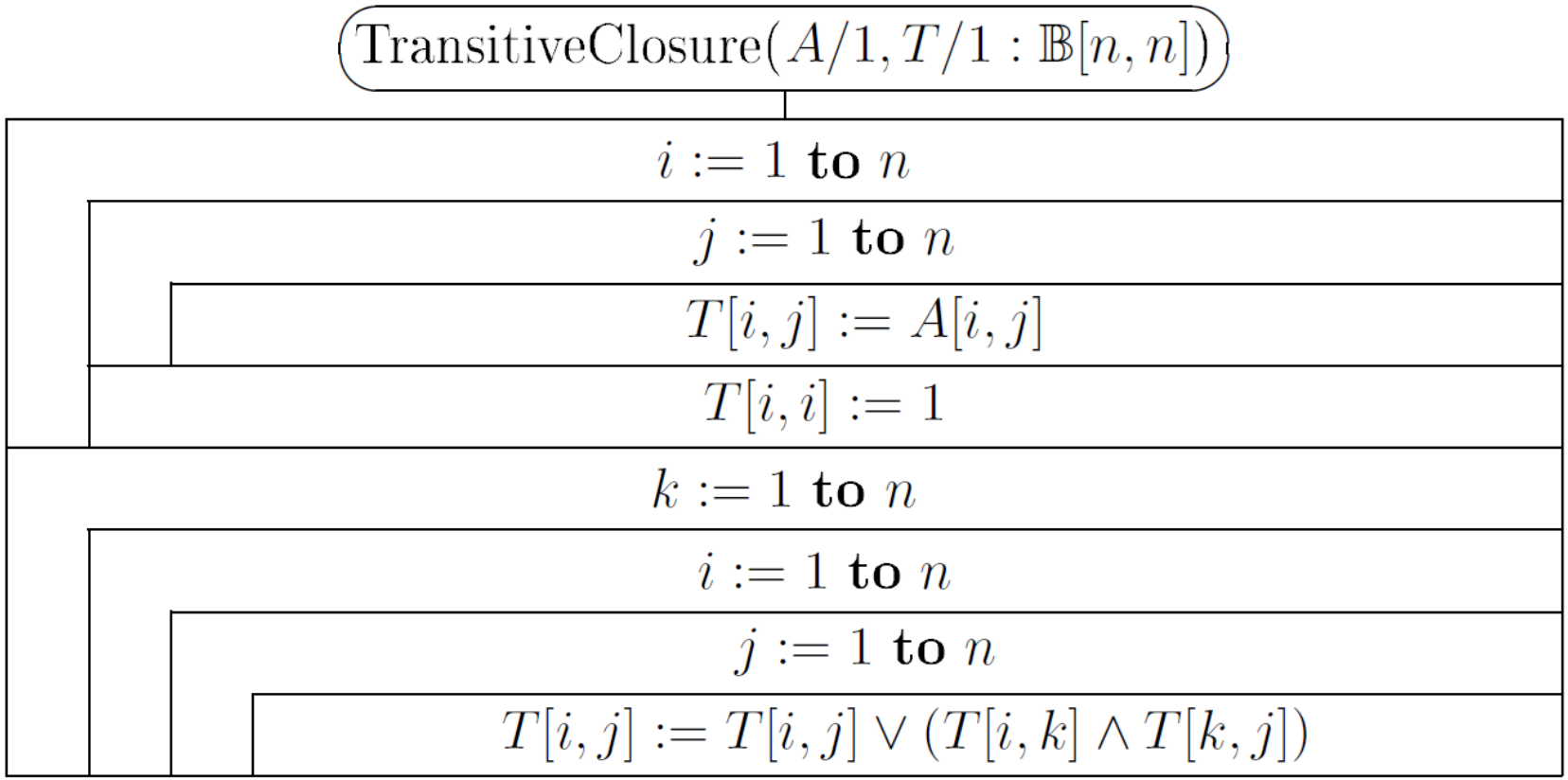
Observation: From the recursive relations we have:

- $T^{(k)}[i,k] = T^{(k-1)}[i,k]$
- $T^{(k)}[k,j] = T^{(k-1)}[k,j]$

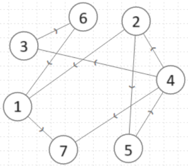
This means that column k and row k of matrix $T^{(k)}$ are the same as the appropriate column and row of matrix $T^{(k-1)}$.

From this it follows that a single matrix T is enough for the whole computation, because $T^{(k)}[i,j]$ depends only on $T^{(k-1)}[i,j]$, $T^{(k-1)}[i,k]$, $T^{(k-1)}[k,j]$.

So this algorithm will also calculate the transitive closure:



Clearly the time complexity for this algorithm is $\in \Theta(n^3)$, space complexity is $\in \Theta(n^3)$.



Directed Graph

	1	2	3	4	5	6	7
1							
2							
3							
4							
5							
6							
7							

	1	2	3	4	5	6	7
1							
2							
3							
4							
5							
6							
7							

	1	2	3	4	5	6	7
1							
2							
3							
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6							
7							

	1	2	3	4	5	6	7
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	1	2	3	4	5	6	7
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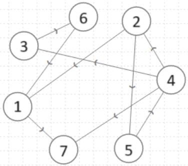
6							
7							



Directed Graph

	1	2	3	4	5	6	7
1							
2							
3							
4							
5							
6							
7							

	1	2	3	4	5	6	7
1							
2							
3							
4							
5							
6							
7							

	1	2	3	4	5	6	7
1							
2							
3							
4							
5							
6							
7							

	1	2	3	4	5	6	7
1							
2							
3							
4							
5							
6							
7							

	1	2	3	4	5	6	7
1	1	0	0	0	0	0	1
2	1	1	1	1	1	1	1
3	1	0	1	0	0	1	1
4	1	1	1	1	1	1	1
5	1	1	1	1	1	1	1
6	1	0	0	0	0	1	1
7	0	0	0	0	0	0	1

Transitive Closure